Syed Muhammad Saad Hussain Zaidi

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EDUCATION

NED University OF Engineering and Technology — Karachi, Pakistan
Bachelor of Science (Applied Physics) – Undergraduate — **October 2022-December 2026**

IJK Tech Hub — Karachi, Pakistan
Applied Data Science with Machine Learning — **October 2023 – November 2024**

SKILLS SUMMARY

- **Languages:** Python, HTML, SQL
- **Frameworks:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Ultralytics (YOLOv8).
- **Tools:** Power BI, Excel, PowerPoint, MySQL, SQLite, Roboflow.
- **Platforms:** Jupyter Notebook, Visual Studio Code, Google Colab.
- **Soft Skills:** Adaptability, Emotional Intelligence, Creativity.
- **Technical Skills:** Statistics, Machine Learning, Deep Learning, Content Writing, Analytics, Data Wrangling, Data Visualization, Object Detection, Image Processing.

WORK EXPERIENCE

DATA ANALYST INTERN | PLACEMENT DOST | PINNACLE FULL-STACK INTERNS |NED University |SUPARCO — **April 2024 - August 2025**

- **Data Analysis and Visualization:** Utilized Power BI to create interactive dashboards and conducted in depth data analysis on large datasets.
- **Machine Learning:** Developed predictive models using Python and machine learning algorithms for various applications.
- **Deep Learning:** Developed and trained various neural network architectures for computer vision, object detection, and predictive modeling applications.
- **Technical Proficiency:** Demonstrated strong programming skills in Python and proficiency in applying machine learning techniques.
- **Communication and Collaboration:** Effectively communicated insights to stakeholders and collaborated with cross-functional teams.
- **Problem Solving:** Applied critical thinking and problem-solving skills to address business challenges through data-driven solutions.

PROJECTS

Solar Panel Detection & Change Detection AI Model – RESOLVE South, SUPARCO Complex, Karachi: LINK — **June 2025 - August 2025**

- Developed and optimized a YOLOv8 based machine learning model to detect and monitor solar panels from satellite imagery, performing model comparison to ensure highest detection accuracy and providing actionable insights for renewable energy monitoring.

Bankruptcy Risk Prediction Using Hybrid Machine Learning Models – NED University: LINK — **February 2025 - June 2025**

- Developed a hybrid Dynamic Panel Probit and ANN model using financial ratios, macroeconomic, and market variables to predict bankruptcy risk in non financial firms, implementing a single hidden layer ANN with 11 neurons, optimized via cross entropy loss and k fold cross validation, achieving high predictive accuracy while outperforming traditional models.

Flood Damage Prediction Model using Machine Learning – NED University: LINK — **December 2024 - January 2025**

- Developed a machine learning pipeline to predict flood-induced structural damage using inundation depth and material type. Applied feature engineering with one-hot encoding and implemented a two-stage classification approach, optimizing Random Forest to achieve 95% accuracy for minor damage prediction and 83% for severe damage classification.

Flight Activity Dashboard with Power BI - Placement Dost: LINK — **April 2024 - June 2024**

- Developed an interactive dashboard to analyze flight booking trends, customer segmentation by CLV, geographical distribution, and loyalty card impact, providing actionable insights for business strategies.

Email Spam Detection System with Python - Pinnacle Full-Stack Interns: LINK — **June 2024 - July 2024**

- Developed a Python-based system for email spam detection using text preprocessing and classification models (e.g., decision trees), enhancing skills in text classification and spam detection techniques.

Heart disease using classification algorithms – Pinnacle- Full-Stack Interns: LINK — **June 2024 - July 2024**

- Created a predictive model using machine learning algorithms to classify heart disease, enhancing understanding of healthcare analytics and predictive modeling for medical diagnosis.

CERTIFICATES

Fundamentals of Python | CERTIFICATE — **November 2023**

- Completed a comprehensive course on Python programming, covering essential concepts such as data types, control structures, functions, and object-oriented programming. Developed a strong foundation in Python, enabling the creation of efficient, scalable code for various applications.

Data Manipulation and Analysis | CERTIFICATE — **January 2024**

- Gained expertise in manipulating and analyzing data using tools like Pandas and NumPy. The course emphasized data cleaning, transformation, and statistical analysis techniques, equipping me with the skills to handle large datasets and extract meaningful insights.

SQL | CERTIFICATE — **February 2024**

- Mastered SQL for querying, managing, and analyzing relational databases. The course included complex joins, subqueries, and performance optimization techniques, enhancing my ability to retrieve and manipulate data effectively for analytical purposes.

Data Visualization | CERTIFICATE — **April 2024**

- Specialized in data visualization techniques using tools like Matplotlib, Seaborn, Power BI, and Excel. Learned to create compelling and informative visual representations of data, aiding in the communication of insights to stakeholders and supporting data-driven decision-making.

Statistics for Data Science | CERTIFICATE — **July 2024**

- Acquired in-depth knowledge of statistical methods, including probability theory, descriptive and inferential statistics, hypothesis testing, and regression analysis. Developed strong analytical skills to interpret data patterns and support data-driven decision-making.

Machine Learning | CERTIFICATE — **November 2024**

- Gained expertise in machine learning algorithms such as supervised and unsupervised learning, decision trees, random forests, and neural networks. Hands-on experience in implementing models using libraries like Scikit-learn and TensorFlow, optimizing predictive analytics, and solving real-world problems.